

L1a: Course Orientation, Markets, and Financial Decisions

CHEME 5660 Cornell University

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we answer how this course works, why financial markets matter even if you never become a trader, and what makes one investment preferable to another.

Today's concepts

- **Course roadmap:** we'll explore three asset classes and combine them into strategies, making and testing financial decisions.
- **Why markets matter:** financial markets have societal impact. Most adults and Cornell itself are investors, while companies use public markets to raise money which funds the development of new products and services.
- **Our first decision problem:** today, we'll introduce our first decision problem. Investments pay different amounts at different times. How can we compare them?

How This Course Runs

Across twenty-eight lectures, we'll move from understanding financial assets to valuing them, combining them into portfolios, and making investment decisions.

MATERIALS

Download each week's materials from the repository Releases page, or clone the repository. Released files stay frozen; corrections are announced.

TOOLS

Use Julia 1.12 through `juliaup`, Anaconda for Jupyter, and VS Code with the Julia extension. See the README for details.

QUESTIONS

Ask questions on Ed Discussion rather than by email, so each answer reaches everyone with the same question.

Final project: automated trading with live data and live market access through the [Alpaca Markets application programming interface](#).

Policies: [grading](#), [attendance](#), [academic integrity](#), and [accommodations](#) are posted in full

Markets Connect You, Cornell, and Companies

YOUR SIDE (2025)

61 %

of U.S. adults had a retirement account.

How the account is invested determines the risk-reward trade-off.

I'm a student, I don't have a retirement account yet.

CORNELL'S SIDE (2025)

\$10.9B

FY2025 endowment

\$498M mainly from the endowment payout went to operations.

About **29% (\$3.16B)** of Cornell's endowment supported financial aid.

COMPANY SIDE

315

corporate IPOs in U.S. markets
01/2025 - 06/2026

An IPO lists a private company's shares on a public market.

Raise capital by selling new shares; **create liquidity** by letting shareholders trade.

Sources: [Fed SHED, 2025](#); [Cornell FY2025 report](#); [Cornell activities](#); [SEC IPO data](#)

U.S. Markets Move About \$1.85 Trillion a Day

One U.S. trading day (\$1.85T) moves about 2.2 times the combined full-year revenue of global biopharma and batteries (\$847B).

	Annual total (USD)	Per trading day (USD)
Global biopharma	666B	2.64B
Global batteries	181B	0.72B
Together	847B	3.36B
U.S. debt	304.7T	1.21T
U.S. stocks	153.2T	608B
Prices paid for options	9.3T	37B
U.S. markets together	467.2T	1.85T

On the same daily basis, U.S. markets move \$1.85T versus \$3.36B in global biopharma and battery revenue, about 550 times as much.

Sources: market sizes 2025 (Fortune Business Insights); turnover 2024--2026 (SIFMA, Cboe)

Each \$1 of Option Premium Corresponds to About \$108 of Gross Notional

In 2025, 60.4 million option contracts were traded each day. Buyers paid USD 37B in premiums for about USD 4T in gross notional value:

$$\frac{\text{USD 4T}}{\text{gross notional value}} \div \frac{\text{USD 37B}}{\text{premium}} = 108\times$$

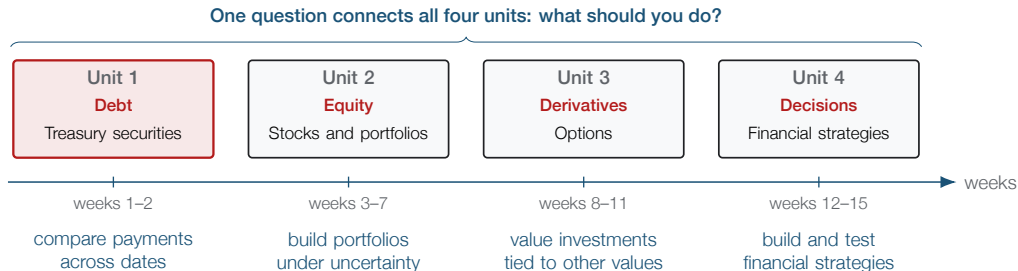
\$108 gross notional per \$1 paid

About USD 3T of that value came from cash-settled S&P 500 index options. For an option, gross notional divided by premium approximates exposure-based leverage when $|\delta| \approx 1$.

Contract size: equity options usually cover 100 shares. For index options, the multiplier converts index points into dollars (SPX: \$100 per point).

The Course Moves from Valuing Assets to Making Decisions

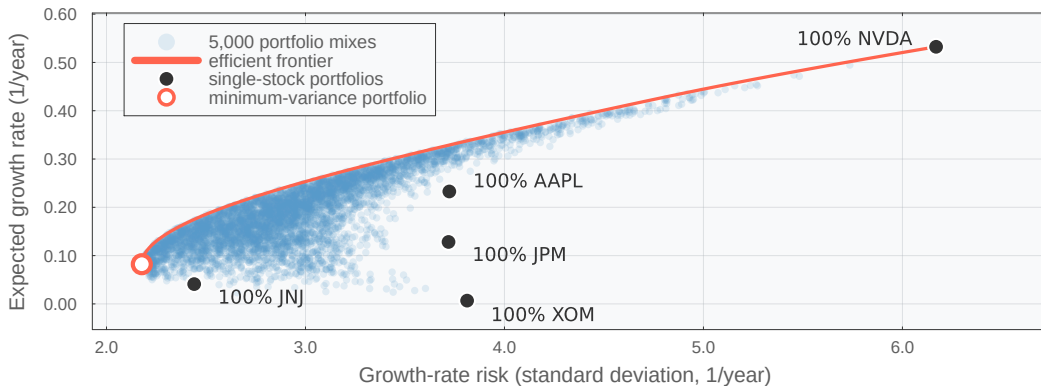
The first three units explore how to value debt, equity, and derivatives. The fourth unit uses these assets to build and test financial strategies.



This week starts with debt in the form of United States Treasury securities.

Optimization Identifies the Risk-Reward Frontier

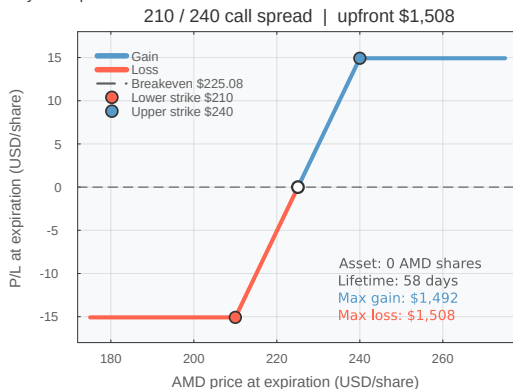
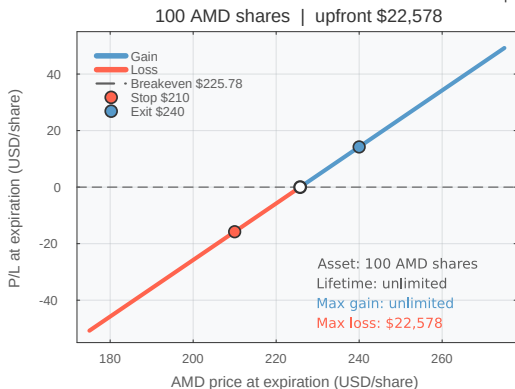
Five stocks—AAPL, JNJ, JPM, NVDA, and XOM—create thousands of portfolios. The red frontier gives the highest expected growth rate at each level of risk.



Options Provide Upside with Limited Downside

At the same \$210/\$240 thresholds, the spread requires \$1,508 upfront instead of \$22,578. Its loss limit is contractual; a stock stop is not.

AMD | 58 days to expiration



Federal Deficits Require Treasury Borrowing

Treasury securities do two jobs: they finance federal deficits and serve as the benchmark “risk-free” asset in U.S. dollar markets.

FEDERAL DEBT | AUG. 20, 2026

\$40.03T

about **\$117,000 per U.S. resident**

Held by the public **\$32.28T**

Intragovernmental **\$7.75T**

LIQUIDITY-SUPPORT BUYBACK | AUG. 20

Securities offered **\$10.16B**

Announced maximum **\$4.00B**

Treasury accepted **\$1.86B**

Treasury retired portions of three older issues,
0.006% of publicly held debt, to support liquidity.

Why this matters: Treasury yields help set borrowing costs and provide the baseline against which other investments are priced.

Note: Scale comparison, not a bill. Sources: [Treasury debt](#); [buyback](#); [Census population](#).

How Should We Compare Future Payments?

Suppose five debt instruments each cost \$10,000 today. Their payment schedules differ in amount, timing, and uncertainty.

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- A** \$10,028 back, four weeks from today
 - B** \$236 every six months for 10 years, then \$10,000
 - C** \$264 every six months for 30 years, then \$10,000
 - D** \$271 every six months for 10 years, then \$10,000
 - E** Fixed-rate coupons on inflation-adjusted principal for 10 years, then the adjusted principal
-

Which promise should we choose? Let's brainstorm and discuss.

Rates: published rates for 2026-08-18, each scaled to a \$10,000 outlay

Each Payment Schedule Represents a Different Security

	What it is called	Market yield
A	4-week Treasury bill	3.63%
B	10-year Treasury note	4.71%
C	30-year Treasury bond	5.28%
D	investment-grade corporate bonds	5.42%
E	10-year inflation-indexed Treasury (TIPS)	2.41% real

Compare the **amount**, **timing**, and **uncertainty** of the payments. The schedule gives the amount and timing. The security type identifies which sources of uncertainty matter.

Careful: D is a bond index. E's 2.41% is a real market yield, not its coupon; TIPS principal adjusts with CPI.

Risk Shapes the Value of Promised Payments

Promised cash flows are only the starting point. Their value also depends on when we may need the money and which risks we bear.

- **C Interest-rate risk.** Selling before maturity exposes the investor to market-price changes; when yields rise, the bond's price falls.
- **D Credit and liquidity risk.** The 0.71-percentage-point premium over the ten-year Treasury compensates investors for these additional risks.
- **E Inflation protection.** Because TIPS principal adjusts with CPI, its 2.41% real yield measures purchasing-power growth rather than nominal dollars.

A sound comparison considers both the payments and the risks that determine their value.

Silicon Valley Bank (SVB) Needed Cash to Cover Withdrawals

\$200B

assets in March 2023

Depositors became concerned and began withdrawing money.

SVB needed cash, so it sold long-term securities before maturity.

Interest rates had risen, so those securities were worth less than SVB had paid.

Interest-rate risk became a liquidity problem.

Clip: [PBS NewsHour, 28 March 2023, 6 minutes](#)

Summary

Today we established why the course matters and introduced our first investment decision.

Key takeaways

- **The course moves from assets to decisions.** We value debt, equity, and derivatives, then combine and test them in financial strategies.
- **Markets connect savings to investment.** Households, Cornell, companies, and the Treasury use markets to save, invest, raise funds, or finance deficits.
- **A payment schedule is only the starting point.** Amount, timing, interest-rate, credit, liquidity, and inflation risk all affect an investment's value.

Next, we build our first comparison rule: use the time value of money to compare payments made on different dates.

Further Reading

These readings are optional and will not appear on a problem set. Use them if you want more background on the Federal Reserve, monetary policy, or the Treasury.

- [*The Structure and Functions of the Federal Reserve System*](#), Federal Reserve Board: who does what inside the Federal Reserve.
- [*Economy at a Glance: Policy Rate*](#), Federal Reserve Board: how the Fed reports and carries out interest-rate policy.
- [*Fedpoints: Federal Reserve System*](#), Federal Reserve Bank of New York: a short overview of how the system is organized.
- [*Role of the Treasury*](#), U.S. Department of the Treasury: what the Treasury does and how it differs from the Federal Reserve.